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## MA 652 Stability Theory of Differential Equations

**Course Code:** MA 652

**Course name:** Stability Theory of Differential Equations

Credits: 3-0-0-3

Prerequisite: Basic calculus, Linear algebra

Intended for: MS/PhD

Approval: 8th Senate; OTA

### Preamble:

This course covers the advance theory of ordinary differential equations. It is designed to be taught to Master and Doctoral level students. This course tries to demonstrate to the students to build advance theory of differential equations.

### Course Outline:

This course is about stability of solutions of differential equations. This is an advance level course which requires basic knowledge of linear algebra and calculus. This course starts with fundamental matrix system. The main aim of this course is to introduce students the advance theory of ordinary differential equations.

### Course Modules:

- General solution, singular solution. [2 Lectures]
- Linear autonomous and nonautonomous systems, Fundamental matrix [10 Lectures]
- Floquet theory, Phase plane analysis, Stability and Lyapunov functions [15 Lectures]
- Limit sets, Periodicity and almost periodicity, Bifurcation and Chaos [15 Lectures]

### Text books:

1. T. A. Burton, **Stability & Periodic Solutions of Ordinary & Functional Differential Equations**, Dover publication.
2. Earl E Coddington, **Theory of Ordinary Differential Equations**, Krieger Publishing Company, 1984.

### Reference books:

1. Lawrence Perko, **Differential Equations and Dynamical Systems**, 3rd Edition, Springer, 2001.